Physics Simulation and STRidge Identification Report

> Numerical identification of the streamwise momentum equation from saved velocity and pressure fields.

1. Simulation Parameters

Quantity	Symbol	Value	Unit
Fluid density	rho	1.00000000e+01	kg m^{-3}
Dynamic viscosity	viscosity	1.00000000e+01	Pa s
Kinematic viscosity	nu	1.00000000e+00	m^2 s^{-1}
Characteristic velocity	u_max	1.00000000e+00	m s^{-1}
Characteristic length	D	1.00000000e+00	m
Reynolds number	Re	1.00000000e+00	dimensionless
Body force	Fx_ext	0.00000000e+00	m s^{-2}
Grid points in x	Nx	3.10000000e+01	cells
Grid points in y	Ny	3.10000000e+01	cells
Grid spacing in x	dx	3.22580645e-02	m
Grid spacing in y	dy	3.22580645e-02	m
Time step	dt	1.00000000e-04	s
Numerical threshold	epsilon	7.00000000e-11	-

2. Numerical Discretisation

Quantity	Symbol	Value	Unit
Parameters listed above are the values used by the selected solver.			

3. Data and Pre-processing

Quantity	Value
Loaded field shape (t, x, y)	[(6678, 80, 80), (1802, 80, 80), (2175, 80, 80), (10000, 80, 80)]
Regression field shape (t, x, y)	[(4678, 78, 78), (901, 78, 78), (1088, 78, 78), (8000, 78, 78)]
Regression samples	−8,268,782,793,637,494,784

Quantity	Value
Dictionary terms	8
First saved time index	1
Temporal crop at each boundary	2000 layers
x-boundary crop	1 points
y-boundary crop	1 points

4. Governing Equation Verification

Reference equation

```
u_t =- 1.00000000e+00*uu_x - 1.00000000e+00*vu_y + 1.00000000e+00*u_xx +
1.00000000e+00*u_yy - 1.00000000e-01*p_x
```

Learned equation

```
u_t =2.38343106e+00*uu_x + 3.71408080e+00*vu_y + 9.22373053e-01*u_xx +
1.02369093e+00*u_yy - 9.79729907e-02*p_x + 1.17584552e-02*u_px - 4.02348133e-
02*v_px + 3.91600747e-02*uu_xx
```

Term	Reference coefficient	Learned coefficient	Percentage error
uu_x	-1.00000000e+00	+2.38343106e+00	-3.38343106e+02%
vu_y	-1.00000000e+00	+3.71408080e+00	-4.71408080e+02%
u_xx	+1.00000000e+00	+9.22373053e-01	7.76269471e+00%
u_yy	+1.00000000e+00	+1.02369093e+00	2.36909274e+00%
p_x	-1.00000000e-01	-9.79729907e-02	-2.02700931e+00%
u_px	+0.00000000e+00	+1.17584552e-02	inf%
v_px	+0.00000000e+00	-4.02348133e-02	inf%
uu_xx	+0.00000000e+00	+3.91600747e-02	inf%

5. STRidge Selection and Error Metrics

Metric	Value
Regularisation parameter, λ	1.00000000e-06
Selected tolerance	1.00000000e-02
L0 penalty	1.00000000e-06
Training RMSE	1.04769632e+01

Metric	Value
Validation RMSE	1.04612489e+01

6. Reproducibility Notes

- Input directory: /Users/minnie/Desktop/PhysicsFYP/ns_solver/stridge_model/output_npz/fourmovingwalls_Nx81/run_001, /Users/minnie/Desktop/PhysicsFYP/ns_solver/stridge_model/output_npz/fourmovingwalls_Nx81/run_002, /Users/minnie/Desktop/PhysicsFYP/ns_solver/stridge_model/output_npz/lidcavity_Nx81/run_001, /Users/minnie/Desktop/PhysicsFYP/ns_solver/stridge_model/output_npz/poiseuille_Nx81/run_002
- Fields were loaded starting at saved time index 1.
- Derivatives were calculated using second-order edge-aware finite differences via numpy.gradient.
- The learned equation is compared term-by-term with the supplied reference coefficients.

7. Transverse Momentum Equation

Reference equation

```
v_t =- 1.00000000e+00*uv_x - 1.00000000e+00*vv_y + 1.00000000e+00*v_xx + 1.00000000e+00*v_yy - 1.00000000e-01*p_y
```

Learned equation

```
v_t =- 1.05148138e+00*uv_x - 6.89790716e+00*vv_y + 1.03644919e+00*v_xx + 9.86194654e-01*v_yy - 9.99228337e-02*p_y - 1.21816122e-02*vp_y + 4.90858646e-02*up_y + 4.19806811e-02*vv_yy
```

Term	Reference coefficient	Learned coefficient	Percentage error
uv_x	-1.00000000e+00	-1.05148138e+00	-5.14813844e+00%
vv_y	-1.00000000e+00	-6.89790716e+00	-5.89790716e+02%
v_xx	+1.00000000e+00	+1.03644919e+00	3.64491903e+00%
v_yy	+1.00000000e+00	+9.86194654e-01	1.38053457e+00%
p_y	-1.00000000e-01	-9.99228337e-02	-7.71663454e-02%
vp_y	+0.00000000e+00	-1.21816122e-02	inf%
up_y	+0.00000000e+00	+4.90858646e-02	inf%
vv_yy	+0.00000000e+00	+4.19806811e-02	inf%

V STRidge Selection and Error Metrics

Metric	Value
Selected tolerance	1.00000000e-02
Training RMSE	7.12635771e+00
Validation RMSE	7.08158121e+00